$\mathcal{T}_n$ THPD Hermitian 正定 Toeplitz 矩阵空间 Barbarsco

反射系数 $\varphi: \mathcal{T}_n \to \mathbb{R}_+ \times D^{n-1}$ 复平面的单位圆盘

$R_n \mapsto (P_0, \mu_1, \cdots, \mu_{n-1})$

注：φ 是双射，可逆

P_0 是 R_n 的第一个元素

R_k 是 R_n 的 k 阶主子式

S_k 是 R_{k+1} 删第一行和最后一列

$$\mu_k = (-1)^k \frac{\det S_k}{\det R_k}$$

反射系数化的优势：黎曼度量是对角阵，构造黎曼流形

黎曼度量 $ds^2 = n\frac{dP_0^2}{P_0^2} + \sum_{k=1}^{n-1}(n-k)\left(\frac{|d\mu_k|}{1-|\mu_k|}\right)^2$ 没有交叉项，故度量为对角阵

注：Bregman 度量，Kahler 度量

黎曼距离 $d^2(R_1, R_2) = n\left|\log\left(\frac{P_2}{P_1}\right)\right|^2 + \sum_{k=1}^{n-1}(n-k)\left(\frac{1}{2}\log\frac{1+\left|\frac{\nu_k-\mu_k}{1-\bar{\mu}_k\nu_k}\right|}{1-\left|\frac{\nu_k-\mu_k}{1-\bar{\mu}_k\nu_k}\right|}\right)^2$

测地线 $\gamma(s, R_1, R_2)$，切向量 $\gamma'(0, R_1, R_2) \in T_{R_1}(\mathbb{R}_+^* \times D^{n-1})$

由黎曼均值求 DOA Chahrour

正则化 Burg 算法 每拍估计一个 THPD 矩阵 共 K 拍 $\{R_k\}_{k=1}^K$

Karcher Barycentre 算法求黎曼均值

对于 K 个 THPD 矩阵 $\{R_k\}_{k=1}^K$

$$A_{t+1} = A_t^{\frac{1}{2}} \exp\left\{\varepsilon \sum_{k=1}^{K} \log\left(A_t^{-\frac{1}{2}} R_k A_t^{-\frac{1}{2}}\right)\right\} A_t^{\frac{1}{2}}$$

黎曼距离 $d^2(R_1, R_2) = \left\|\operatorname{logm}\left(R_1^{-\frac{1}{2}} R_2 R_1^{-\frac{1}{2}}\right)\right\|_F^2 = \sum \log^2(\lambda_k)$

λ_k 是 $R_1^{-\frac{1}{2}} R_2 R_1^{-\frac{1}{2}}$ 的特征值

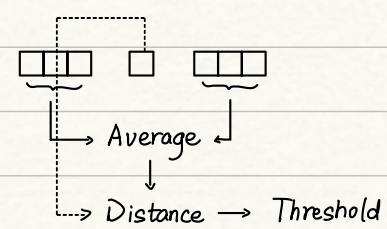
协方差信息几何 Barbarsco

把 $x(t)$ 看作复高斯分布，$x \sim CN(0, R)$

$$p(x|R) = \frac{1}{\pi^M \det(R)} \exp\{-x^H R^{-1} x\}$$

$$p(X|R) = \frac{1}{\pi^M \det(R)} \exp\{-\operatorname{Tr}(\hat{R} R^{-1})\}$$

雷达信号方向判别



由 Fisher 信息作为黎曼度规 $g_{ij}(\theta) = E(\partial_i l \partial_j l)$

面积元 $ds^2 = Tr(R^{-1}dR) = \| R^{-\frac{1}{2}} dR R^{-\frac{1}{2}} \|_F^2$

两个HPD矩阵的自然距离 $d(R_1, R_2)$

$$d(R_1, R_2) = \sum_{i=1}^{n} (\log \lambda_i)^2$$

$\lambda_1 \cdots \lambda_n$ 是 $\det(\lambda R_1 - R_2)$ 的根 也称广义特征值

即 $R_1^{-1}R_2$ 的特征值. 若 R_1 可逆

两个HPD矩阵的测地线

$$\gamma(t) = R_1^{\frac{1}{2}} (R_1^{-\frac{1}{2}} R_2 R_1^{-\frac{1}{2}})^t R_1^{-\frac{1}{2}}, \quad 0 \le t \le 1$$

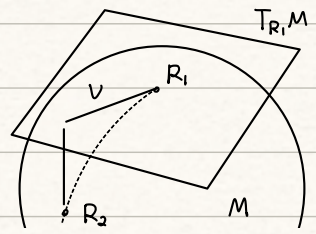
$$\gamma'(t) = R_1^{\frac{1}{2}} (R_1^{-\frac{1}{2}} R_2 R_1^{-\frac{1}{2}})^t \operatorname{logm}(R_1^{-\frac{1}{2}} R_2 R_1^{-\frac{1}{2}}) R_1^{\frac{1}{2}}$$

$$\gamma'(0) = R_1^{\frac{1}{2}} \operatorname{logm}(R_1^{-\frac{1}{2}} R_2 R_1^{-\frac{1}{2}}) R_1^{\frac{1}{2}} \in T_{R_1}M$$

指数映射 $\operatorname{Exp}_{R_1} : T_{R_1}M \to M$

$$V \mapsto R_2$$

$$R_2 = R_1^{\frac{1}{2}} \operatorname{expm}\{R_1^{-\frac{1}{2}} V R_1^{-\frac{1}{2}}\} R_1^{\frac{1}{2}}$$



KL散度 $d^2(R_1, R_2) = Tr(R_2^{-1}R_1) + \log\det \frac{R_2}{R_1} - M$

SPA算法 $d^2(R_1, R_2) = \| R_2^{-\frac{1}{2}} (R_1 - R_2) R_1^{-\frac{1}{2}} \|_F$

Wasserstein 距离

$$d^2(R_1, R_2) = Tr\{R_1 + R_2 - 2(R_1^{\frac{1}{2}} R_2 R_1^{\frac{1}{2}})^{\frac{1}{2}}\}$$

Bures-Wasserstein 度量

Wasserstein 距离

$$\min \quad tr\{\hat{R} + R - 2(\hat{R}^{\frac{1}{2}} R \hat{R}^{\frac{1}{2}})^{\frac{1}{2}}\}$$

$$s.t. \quad R \in T \cap \mathbb{R}_k^{N\times N} \cap \mathcal{H} \cap \mathcal{X}_+$$

$$\min \quad tr\{\hat{R} + R - 2(\hat{R}^{\frac{1}{2}} R \hat{R}^{\frac{1}{2}})^{\frac{1}{2}}\} - \mu \log\det(R) \qquad \text{正定}$$

Toeplitz 投影

$$X = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{pmatrix} \qquad vec(X) = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 1 \\ 2 \\ 3 \\ 2 \\ 1 \end{pmatrix} = S\theta = \begin{pmatrix} 1 & & & & & \\ & 1 & & & 1 & \\ & & 1 & & & \\ & 1 & & & & \\ 1 & & & & & \\ & 1 & & & & \\ & & & 1 & & \\ & 1 & & & & \\ 1 & & & & & \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

线性结构矩阵

$$\mathcal{X}_S = \{ X \in \mathbb{C}^{n\times n} . \ \underset{np\times 1}{vec(X)} = \underset{np\times q}{S}\underset{q\times 1}{\theta}, \ \theta \in \mathbb{C}^n \}$$

$S_{n^2\times n}$ 是根据 Toeplitz 矩阵维数设计的

任意给一个矩阵 A，可求出向 Toeplitz 线性结构的投影 $\bar{X} = P_{\mathcal{X}_S}(A)$

$$\bar{\theta} = S^{\dagger} vec(A), \quad \bar{X} = mat_{n,n}(S\bar{\theta})$$

Hankel 矩阵

X 参数化 $\underset{r^2\times 1}{vec(X)} = \underset{r^2\times 2r}{S}\theta = \underset{r^2\times 2r}{S}\underset{2r\times 1}{g(a,b)}$

$$\text{友矩阵 } \underset{r\times r}{\Phi} = \begin{pmatrix} -a_1 & -a_2 & \cdots & -a_{r-1} & -a_r \\ 1 & 0 & & 0 & \\ 0 & 1 & & 0 & \vdots \\ & & \ddots & & \\ 0 & 0 & & 1 & 0 \end{pmatrix} \qquad [X]_{ij} = \underset{1\times r}{b^T} \underset{r\times r}{\Phi^{i+j-2}} \underset{r\times 1}{e_1}$$

固定秩

SVD 得 $A = U\Sigma V$，低秩投影 $\bar{X} = U\,diag(\sigma_1 \cdots \sigma_r, 0 \cdots 0)\,V$

近端

$$\min \quad f(x) + \frac{1}{2c}\|x - x_k\|^2 \qquad f(x) \text{ 凸}$$

交替

$$\min \quad f(x,y) \qquad \text{凸}$$

$$s.t. \quad x \in \mathcal{X}, \ y \in \mathcal{Y} \qquad \text{非凸优化}$$